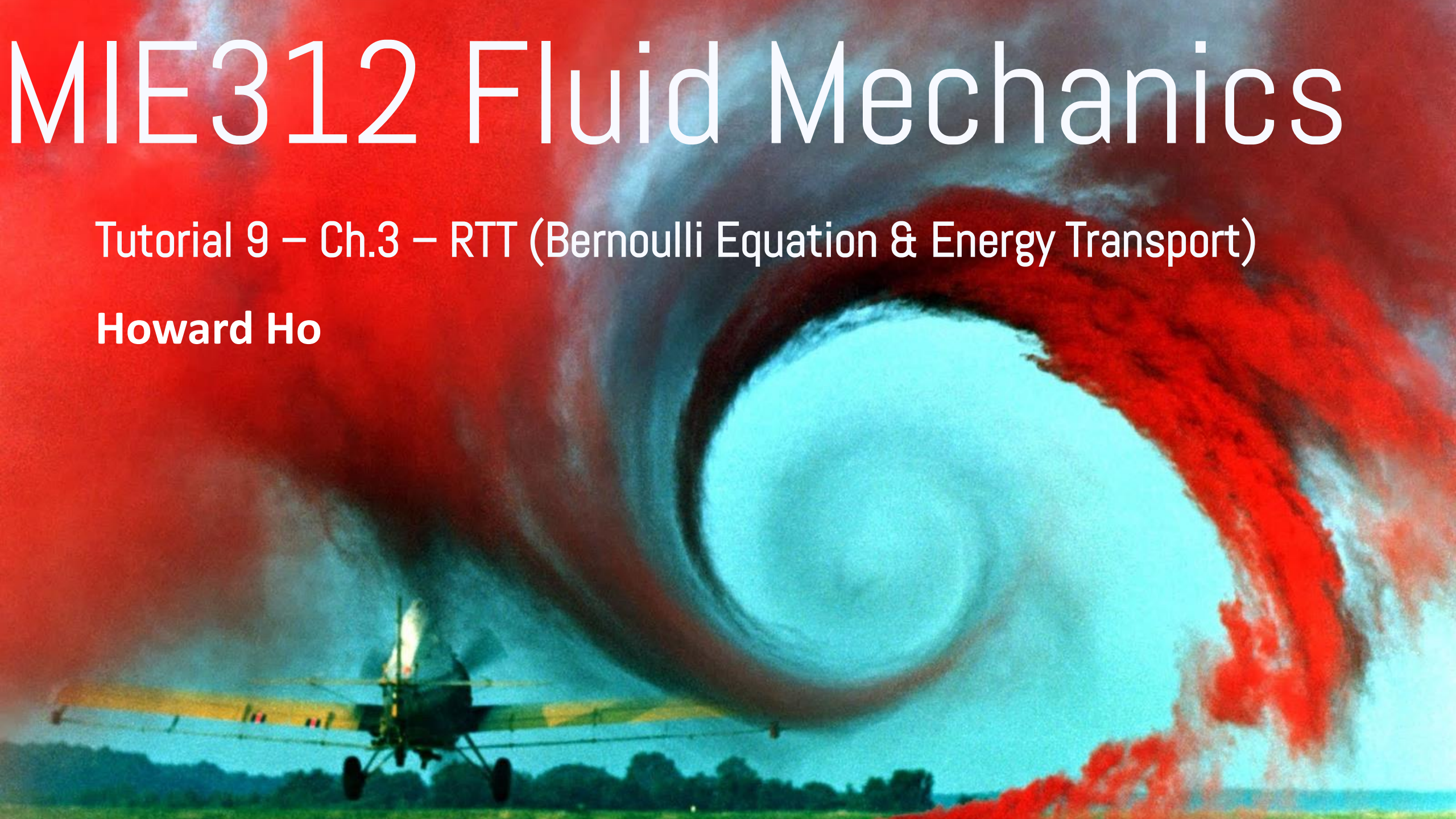
MIE312 Fluid Mechanics

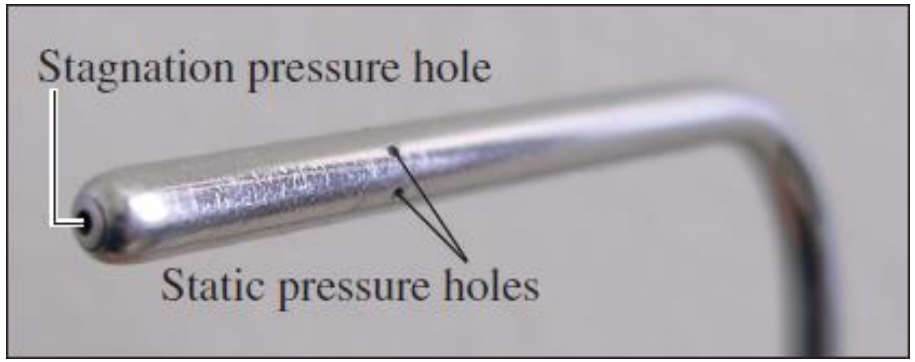
Tutorial 9 – Ch.3 – RTT (Bernoulli Equation & Energy Transport)

Howard Ho

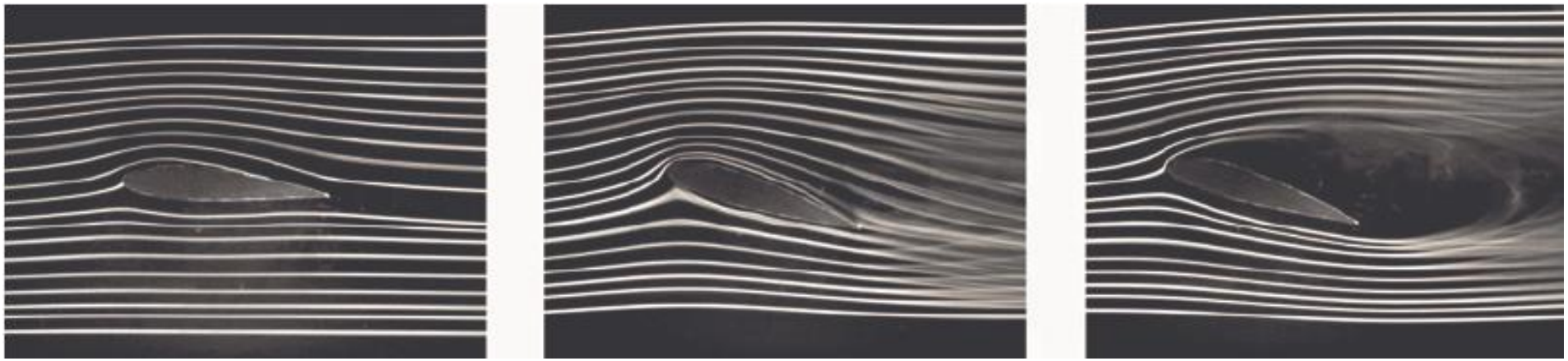


Learning Objective

- Identify when Bernoulli Equation is valid
- Applying the Bernoulli Equation
- Solving problem with Energy Transport Equation (RTT)







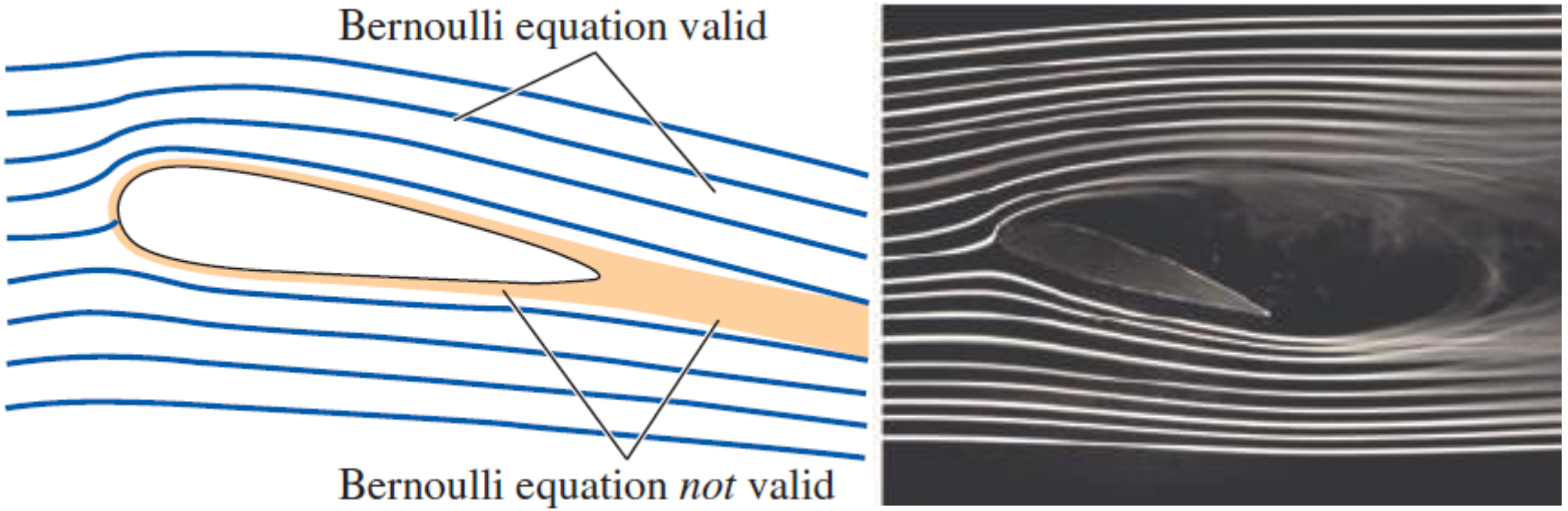
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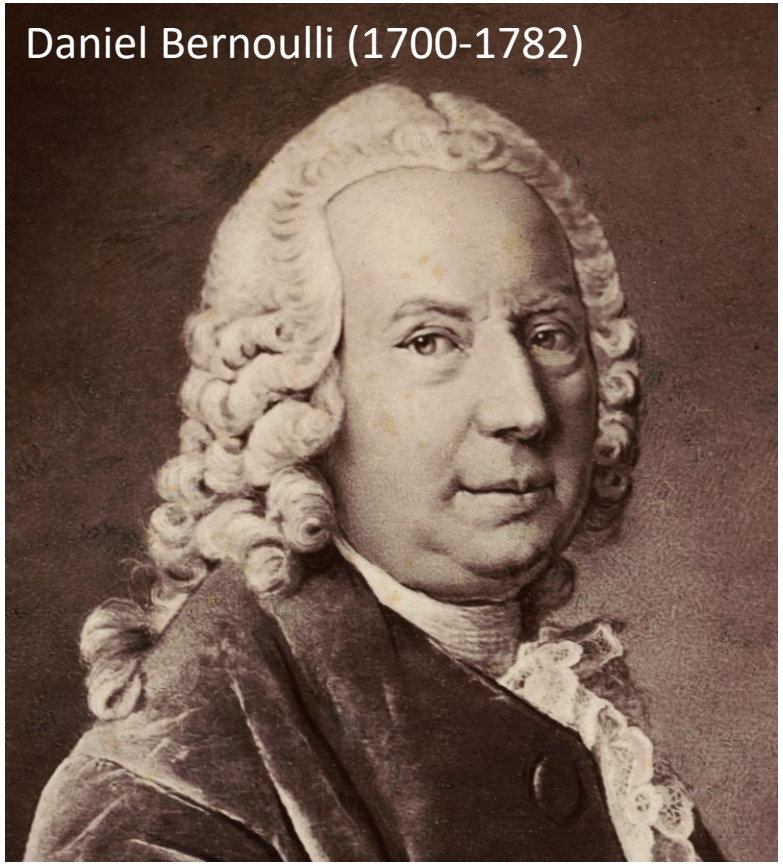
The Bernoulli Equation

- Approximates relation between pressure, velocity, and elevation
- Valid in regions of steady, incompressible flow where net frictional forces are negligible.

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = \text{constant}$$

Despite its simplicity, it has proven to be a very powerful tool in fluid mechanics, but also one of the **most abused equation** in Fluid Mechanics.

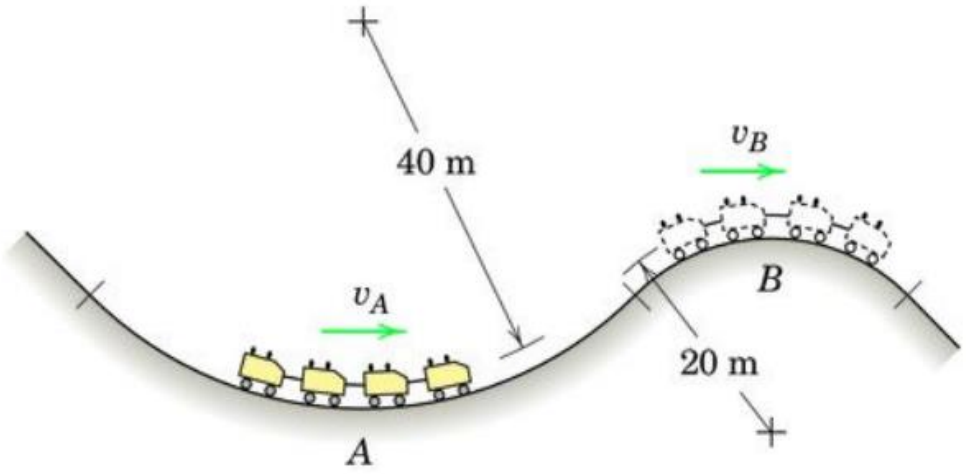




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Conservation of Energy

Solid

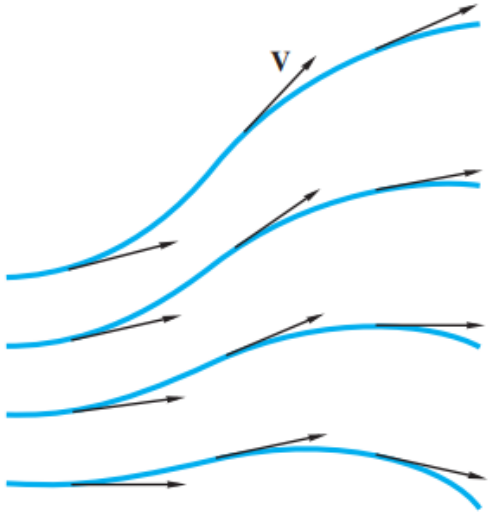


$$PE_A + KE_A = PE_B + KE_B$$

$$mgh_A + \frac{1}{2}mV_A^2 = mgh_B + \frac{1}{2}mV_B^2$$

The Bernoulli equation states that *the total pressure along a streamline is constant*.

Fluid



$$\frac{P_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \frac{V_2^2}{2} + gz_2$$

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

Static Pressure

Dynamic Pressure

Hydrostatic Pressure

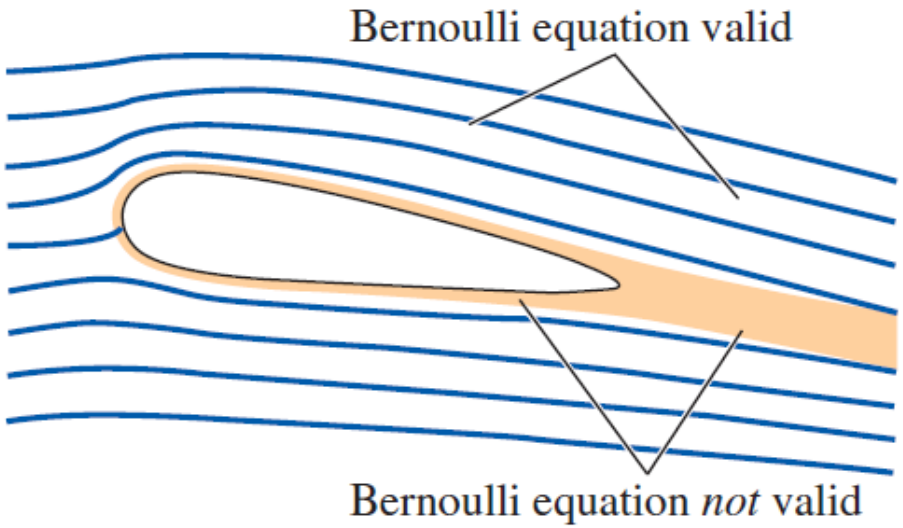
Total Pressure

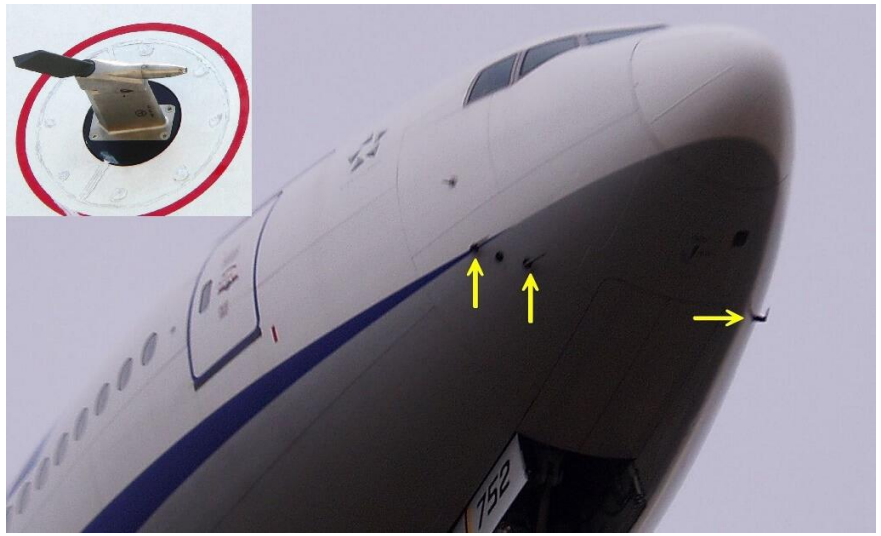
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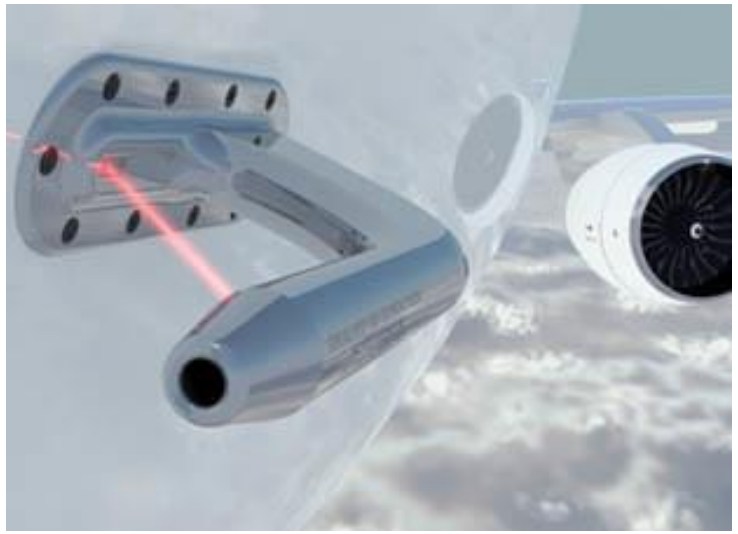
Stagnation Point

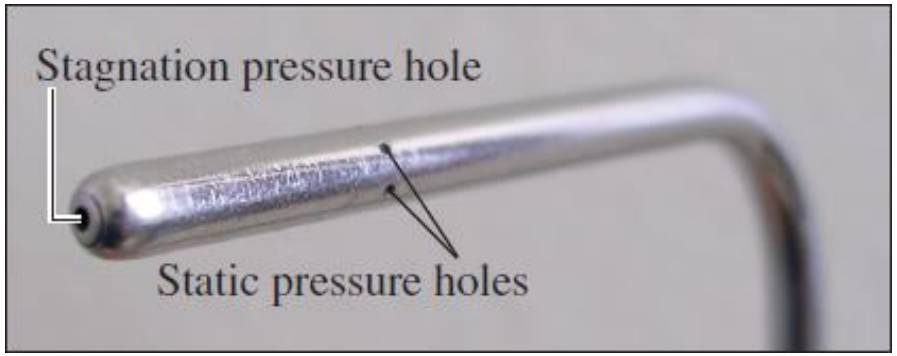
- Stagnation point is the location where the flow is brought to a stop
- Quite often this is located near the nose of an aircraft

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$









Close-up of a **Pitot-static probe**, showing the stagnation pressure hole and two of the five static circumferential pressure holes.

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Stagnation Pressure

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = \text{constant (along a streamline)}$$

Static Pressure

Dynamic Pressure

Hydrostatic Pressure

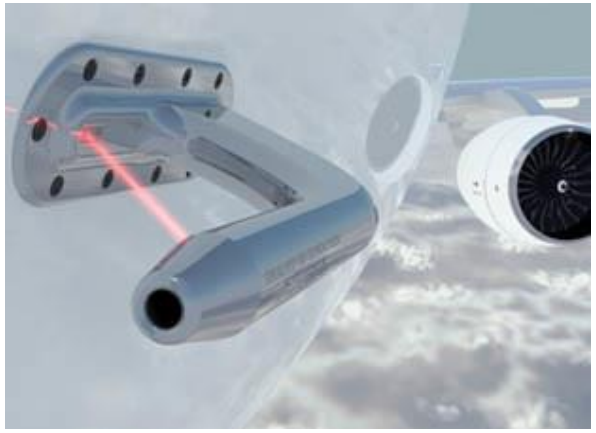
Stagnation pressure: The sum of the static and dynamic pressures. It represents the pressure at a point where the fluid is brought to a complete stop isotropically (in all direction).

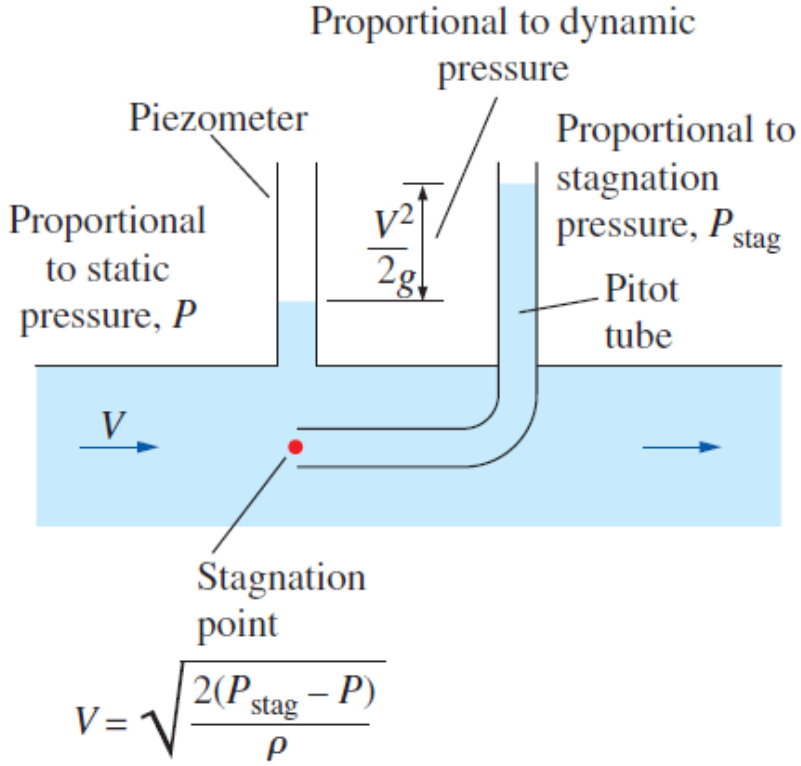
$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

$$P_1 + \frac{1}{2}\rho V_1^2 = P_{stag}$$

Let's say P_2 is the stagnation point and $z_1 = z_2$

$$V_1 = \sqrt{\frac{2(P_{stag} - P_1)}{\rho}}$$





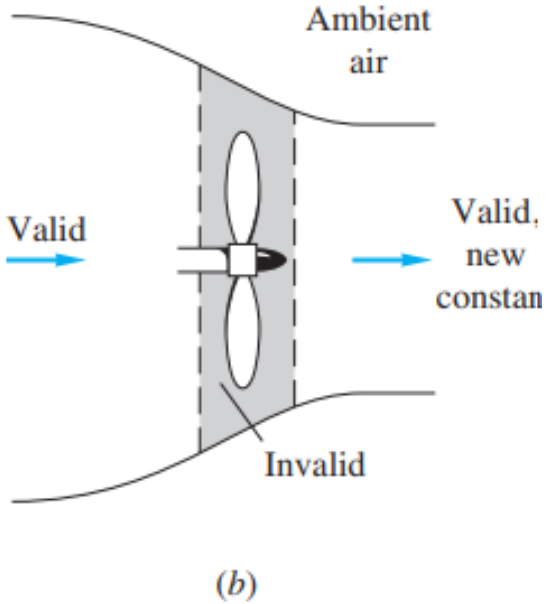
The static, dynamic, and stagnation pressures measured using **piezometer tubes**.

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Limitations of the Bernoulli Equation

5 Steps Guide (In the context of MIE 312)

1. Steady flow
2. Frictionless flow
3. No shaft work (includes a pump, turbine, fan, or any other machine or impeller)
4. Incompressible flow
5. Flow along a streamline



$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = \text{constant}$$

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Energy Conservation

For **Standard** Bernoulli Equation
1. Steady flow
2. Frictionless flow
3. No shaft work
4. Incompressible flow
5. Flow along a streamline

- Standard Bernoulli Equation:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

- We cannot create or destroy energy, we can only change their form (1st Law of Thermodynamics)

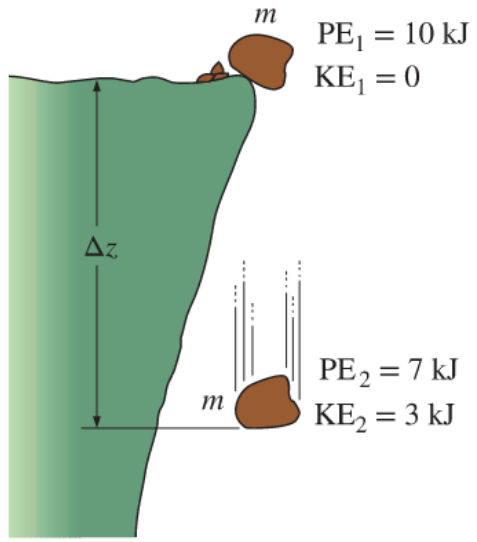
- If we apply this in the form of RTT (Reynolds Transport Theorem):

Q = Heat Transfer
W_s = Shaft Work
W_v = Viscous stress Work (Friction)

$$\dot{Q} - \dot{W}_s - \dot{W}_v = \frac{\partial}{\partial t}\left[\int_{CV}\left(\hat{u} + \tfrac{1}{2}V^2 + gz\right)\rho d\mathcal{V}\right] + \int_{CS}\left(\hat{h} + \tfrac{1}{2}V^2 + gz\right)\rho(\mathbf{V}\cdot\mathbf{n})\,dA$$

$$\dot{Q} - \dot{W}_s - \dot{W}_v = -\dot{m}_1(\hat{h}_1 + \tfrac{1}{2}V_1^2 + gz_1) + \dot{m}_2(\hat{h}_2 + \tfrac{1}{2}V_2^2 + gz_2)$$

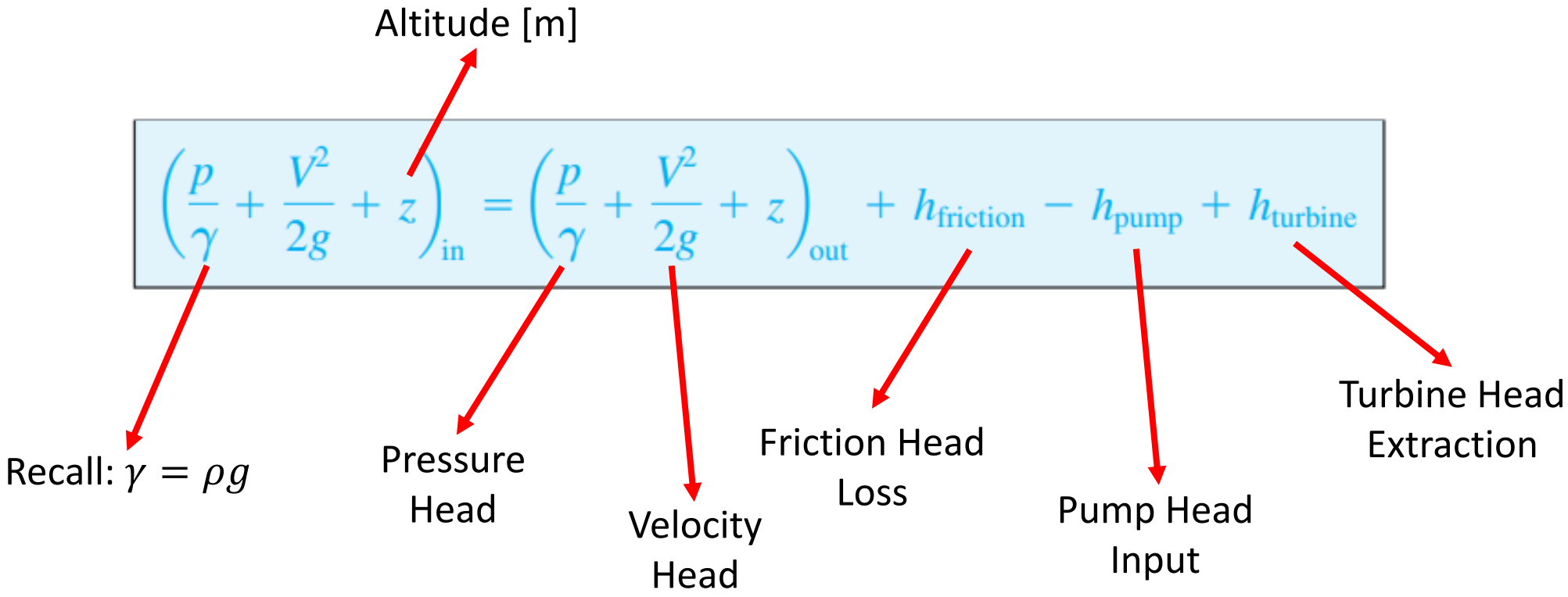
$$\left(\frac{p}{\gamma} + \frac{V^2}{2g} + z\right)_{in} = \left(\frac{p}{\gamma} + \frac{V^2}{2g} + z\right)_{out} + h_{friction} - h_{pump} + h_{turbine}$$



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Energy Equation

- The new equation allows us to look at pump and turbine problems, note it is no longer the Bernoulli Equation (even if they look similar)



Standard Bernoulli Equation:

$$P_1+\frac{1}{2}\rho V_1^2+\rho g z_1=P_2+\frac{1}{2}\rho V_2^2+\rho g z_2$$

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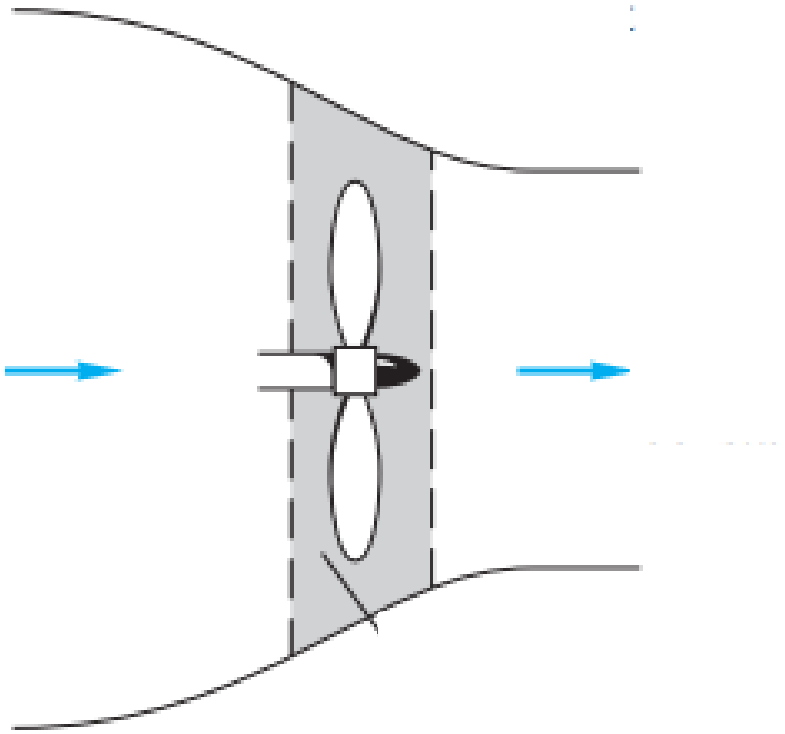
Pump & Turbine System

$$\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{\text{in}}=\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{\text{out}}+h_{\text{friction}}-h_{\text{pump}}+h_{\text{turbine}}$$

- Friction loss is always positive in real (viscous) flows, a pump adds energy (increases the left-hand side), and a turbine extracts energy from the flow.
- Let's say the turbine on the right extract 1 unit of energy

$$\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{\text{in}}=\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{\text{out}}+h_{\text{friction}}-h_{\text{pump}}+h_{\text{turbine}}$$

$$0+2+0=0+1+0+0-0+1$$



- To find Power, simply use: $P=\gamma Q\,h$

e.g. $P_{turbine}=\gamma Q\,h_{turbine}$

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Same Streamline*

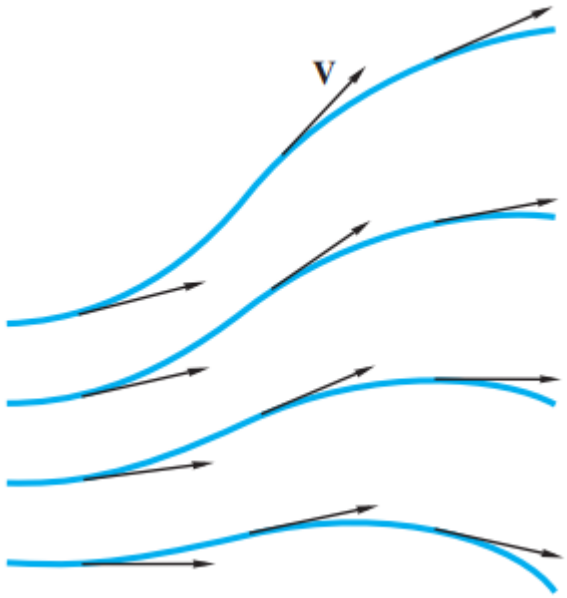
Note: Call back to roller-coaster comparison, we cannot jump from one roller coaster to another

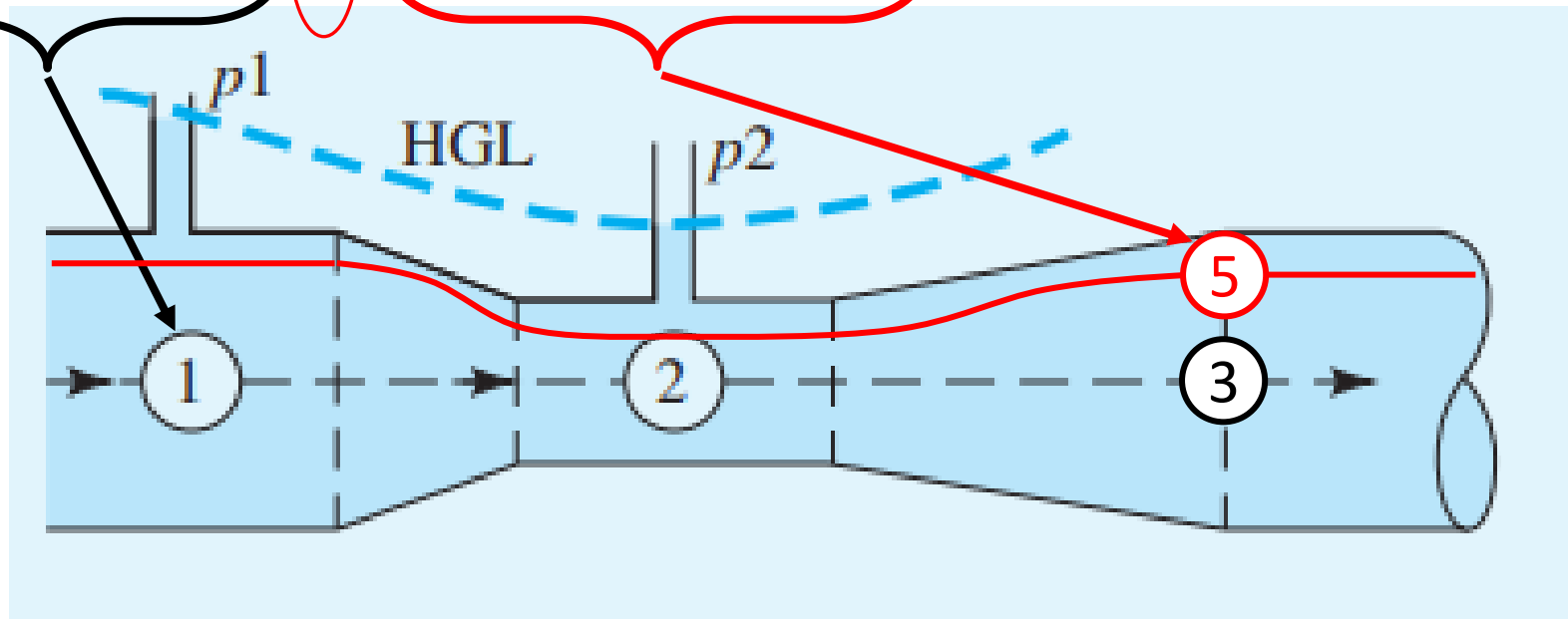
- Valid (same streamline):

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_3 + \frac{1}{2}\rho V_3^2 + \rho g z_3$$

- For an invalid case (not the same streamline) where we try to apply Bernoulli equation across different streamlines, it would be wrong.

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 \neq P_5 + \frac{1}{2}\rho V_5^2 + \rho g z_5$$

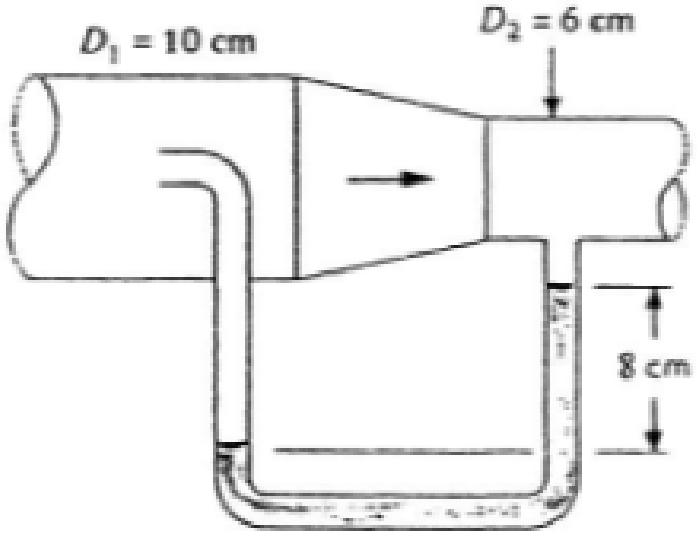




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Problem 1

In the figure on the right, the flowing fluid is CO_2 at $20°C$. Neglect losses. If $p_1 = 170\ kPa$ and the manometer fluid is Meriam red oil (SG = 0.827), estimate $(a) p_2$ and (b) the gas flow rate in m^3/h.



3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}; \quad \dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}; \mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}\,rV_\theta - \sum_{in}\dot{m}\,rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$

Force of Jet on Vane:

$F_x = \dot{m}(V_{1,x} - V_{2,x}); \quad F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

Estimate the CO_2 density as $\rho = p/RT = (170000)/[189(293)] \approx 3.07\ kg/m^3$.

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4. PITOT TUBE & PRESSURE TYPES

Static Pressure (P): Actual thermodynamic pressure.

Stagnation (P_0): Pressure when $V \to 0$ isentropically.

Dynamic Pressure: $q = \frac{1}{2}\rho V^2$

Relation: $P_0 = P + \frac{1}{2}\rho V^2 \to V = \sqrt{\frac{2(P_0 - P)}{\rho}}$

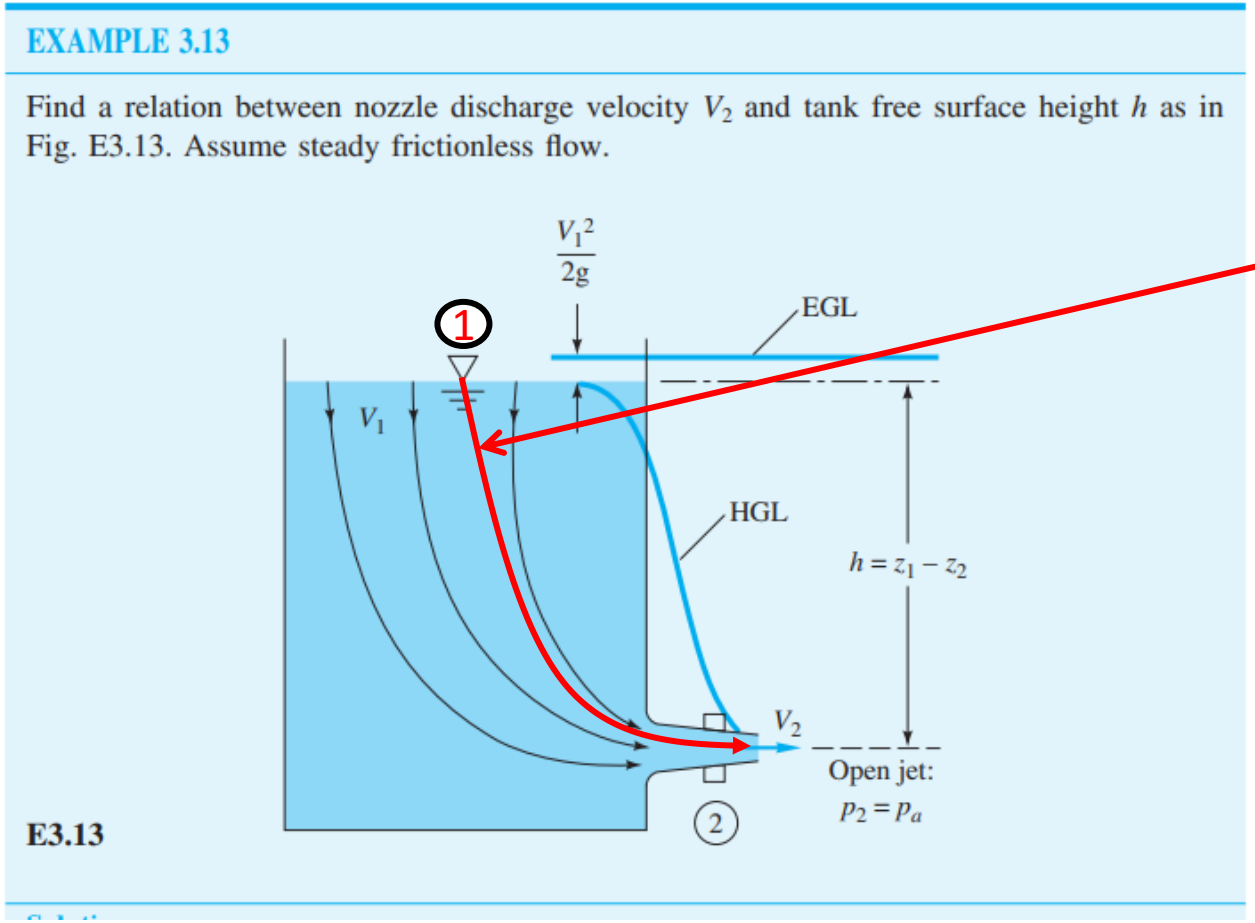
Pitot-Static Tube: $\Delta P = h(\gamma_m - \gamma_f)$

$\to V = \sqrt{2gh(\rho_m/\rho_f - 1)}$

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Reservoir*

- For reservoir tanks, technically we want to make sure we are tracking a streamline from the centre of the tank to the exhaust (although in reality this condition is not very important, if you pick any of the black lines away form the centre it does not change the math).



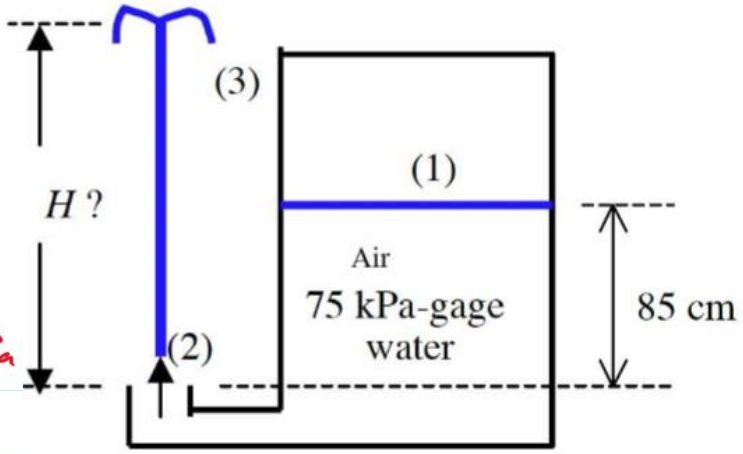
The streamlines in this area doesn't matter much, we can just assume we are always starting from the middle of the tank's water surface (in red).

Modified from Textbook, E3.13

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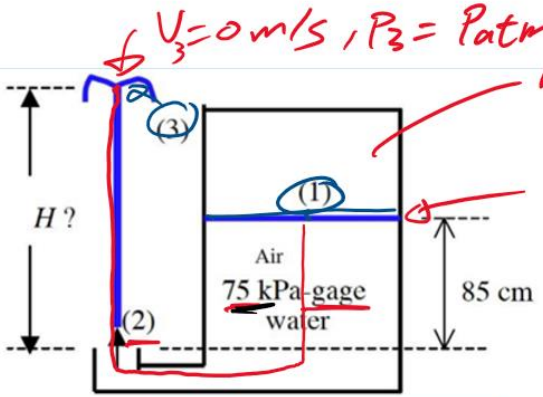
Problem 2* (From 2024)

Water at $20°C$, in the pressurized tank flows out and creates a vertical jet as shown. Assuming steady state frictionless flow, determine the height H to which the jet rises.



14 October 2024 13:39

P3.117 Water at 20°C, in the pressurized tank of Fig. P3.117, flows out and creates a vertical jet as shown. Assuming steady frictionless flow, determine the height H to which the jet rises.



$V_3 = 0$ m/s, $P_3 = P_{atm}$, $P_{3,gage} = 0$ Pa

Reservoir

Set datum @ (2) $\therefore z_2 = 0$

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_3 + \frac{1}{2}\rho V_3^2 + \rho g z_3$$

P_1 (1): $75(1000)$ Pa; z_1 (1): 0.85 m; z_3 (3): H

$$75(1000) + 0 + 998(9.81)(0.85) = 0 + 0 + 998(9.81)H$$

$$H = 8.51 \text{ m}$$

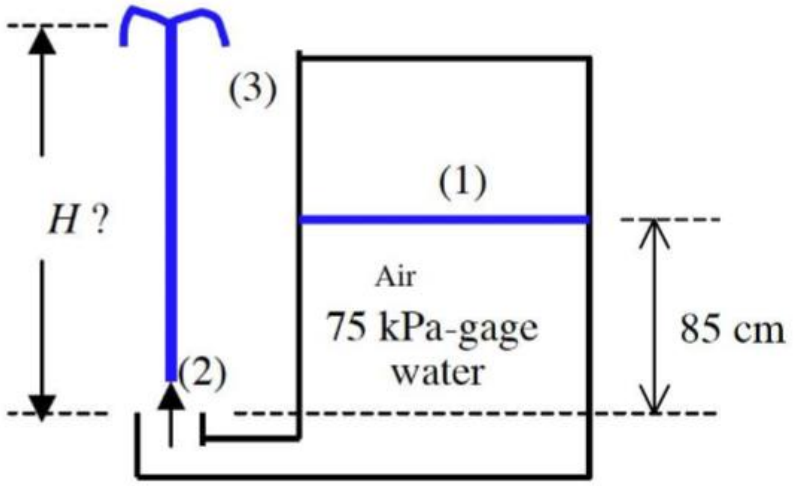
5 Rules for Bernoulli Equation
1. Steady flow ✓
2. Frictionless flow ✓
3. No shaft work ✓
4. Incompressible flow ✓
5. Flow along a streamline ✓

5 Rules for Bernoulli Equation
1. Steady flow
2. Frictionless flow
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Problem 2*

P3.117 Water at 20°C, in the pressurized tank of Fig. P3.117, flows out and creates a vertical jet as shown. Assuming steady frictionless flow, determine the height H to which the jet rises.



Solution: This is a straightforward Bernoulli problem. Let the water surface be (1), the exit plane be (2), and the top of the vertical jet be (3). Let $z_2 = 0$ for convenience.

If we are clever, we can bypass (2) and write Bernoulli directly from (1) to (3):

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_3}{\rho g} + \frac{V_3^2}{2g} + z_3 \quad , \quad or:$$

$$\frac{75000}{(9.81)(998)} + 0 - 0.85m = 0 + 0 + H$$

$$Solve \qquad H = 7.66m + 0.85m = \mathbf{8.51\,m} \qquad Ans.$$

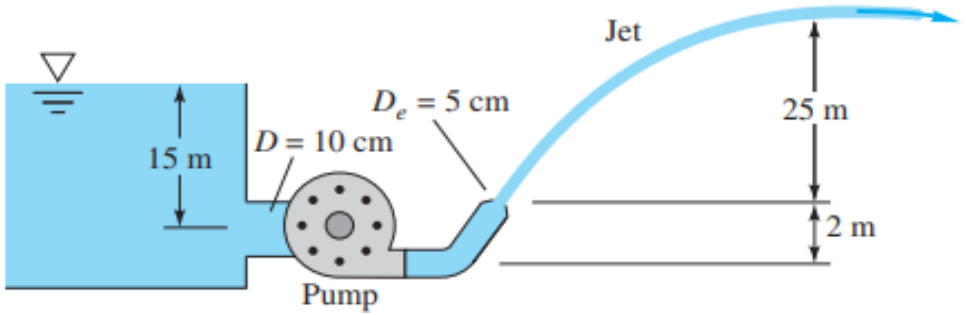
5 Rules for Bernoulli Equation

1. Steady flow
2. Frictionless flow
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5. Flow along a streamline

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Question 3

The pump shown in the figure creates a 20°C water jet with a velocity $v_e = 31.25$ m/s. System friction head losses are 6.5 m. The jet may be approximated by the trajectory of frictionless particles. What power must be delivered by the pump?



3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}$; $\dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}rV_\theta - \sum_{in}\dot{m}rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$

Force of Jet on Vane:

$F_x = \dot{m}(V_{1,x} - V_{2,x})$; $F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

6. INTERNAL PIPE FLOW BASICS

Head Loss: $h_L = (f\frac{L}{D} + \sum K_L)\frac{V^2}{2g}$

Power: $P = \omega T_o$; $P = \gamma Q h$

Pump Efficiency: $\eta = \frac{P_{pump}}{P_{input}}$

Laminar ($Re < 2300$): $f = 64/Re$; $Q = \frac{\pi D^4\Delta P}{128\mu L}$

Entry: $L_{e,lam} \approx 0.06ReD$; $L_{e,turb} \approx 4.4Re^{1/6}D$

Non-Circular: $D_h = 4A/P_w$; **Junctions:** $\sum Q_{node} = 0$

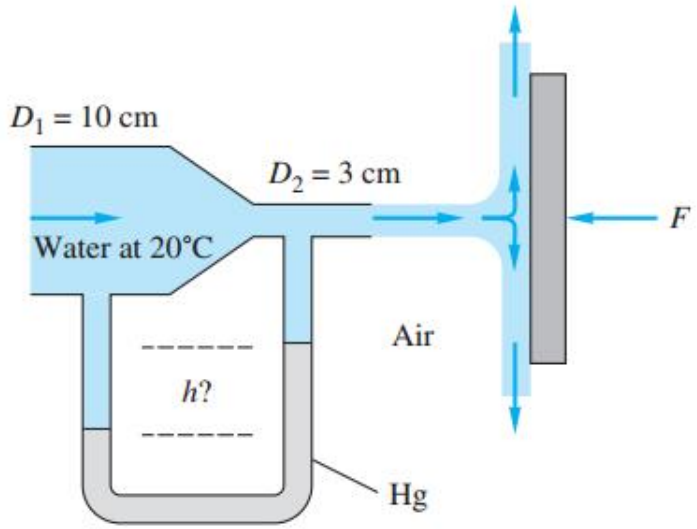
Hardy Cross: $\Delta Q = -\frac{\sum h_L}{2\sum(h_L/Q)}$

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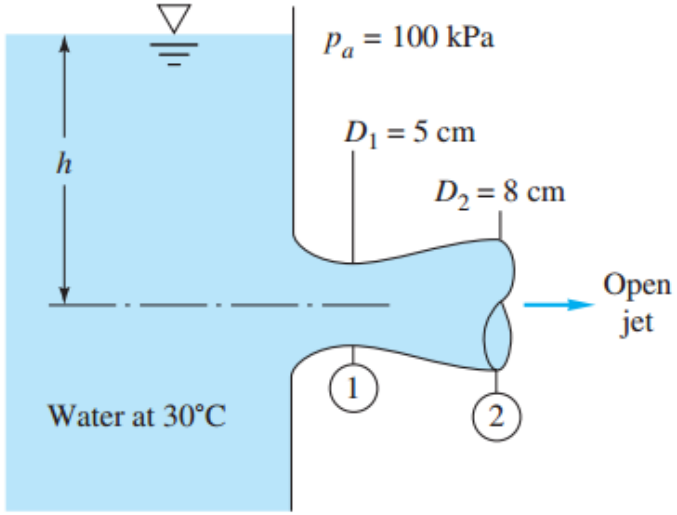
Practice Problems

P3.114 Water flows through a circular nozzle, exits into the air as a jet, and strikes a plate, as shown in Fig. P3.114. The force required to hold the plate steady is 70 N. Assuming steady, frictionless, one-dimensional flow, estimate (*a*) the velocities at sections (1) and (2) and (*b*) the mercury manometer reading *h*.



P3.114

P3.133 If losses are neglected in Fig. P3.133, for what water level *h* will the flow begin to form vapor cavities at the throat of the nozzle?



P3.133

3.126 Given a steady isothermal flow of water at 20°C through the device in Fig. P3.126. Heat-transfer, gravity, and temperature effects are negligible. Known data are $D_1 = 9$ cm, $Q_1 = 220$ m^3/h, $p_1 = 150$ kPa, $D_2 = 7$ cm, $Q_2 = 100$ m^3/h, $p_2 = 225$ kPa, $D_3 = 4$ cm, and $p_3 = 265$ kPa. Compute the rate of shaft work done for this device and its direction.

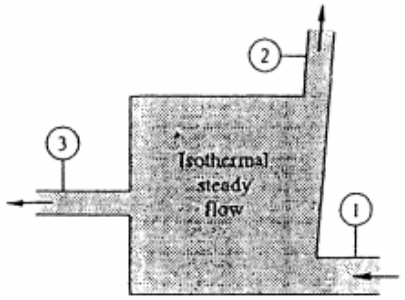


Fig. P3.126

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Practice Problem Solution

P3.114 Water flows through a circular nozzle, exits into the air as a jet, and strikes a plate, as in Fig. P3.114. The force required to hold the plate steady is 70 N. Assuming frictionless one-dimensional flow, estimate (a) the velocities at sections (1) and (2); (b) the mercury manometer reading h.

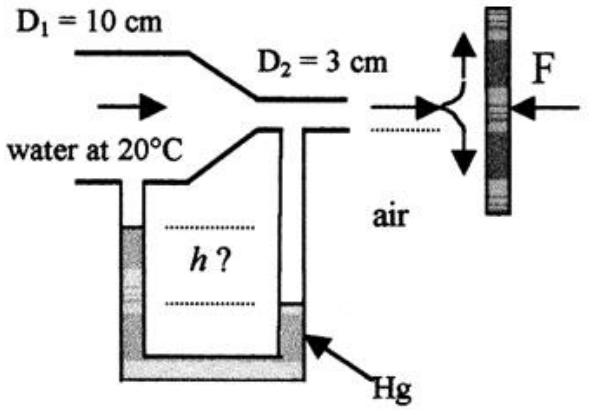


Fig. P3.114

Solution: (a) First examine the momentum of the jet striking the plate,

$$\sum F = F = -\dot{m}_{in}u_{in} = -\rho A_2 V_2^2$$

$$70\ N = -(998)\left(\frac{\pi}{4}\right)(0.03^2)(V_2^2) \quad \boldsymbol{V_2 = 9.96\ m/s} \quad Ans.\ (a)$$

$$Then \quad V_1 = \frac{V_2 A_2}{A_1} = \frac{(9.96)\left(\frac{\pi}{4}\right)(0.03^2)}{\frac{\pi}{4}(0.1^2)} \quad or \quad \boldsymbol{V_1 = 0.9\ m/s} \quad Ans.\ (a)$$

(b) Applying Bernoulli,

$$p_2 - p_1 = (1/2)\rho\left(V_2^2 - V_1^2\right) = (1/2)(998)(9.96^2 - 0.9^2) = 49{,}100\ Pa$$

And from our manometry principles,

$$h = \frac{\Delta p}{\Delta(\rho g)} = \frac{49{,}100}{(133{,}100 - 9790)} \approx \boldsymbol{0.40\ m} \quad Ans.\ (b)$$

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Practice Problem Solution

P3.133 If losses are neglected in Fig. P3.133, for what water level h will the flow begin to form vapor cavities at the throat of the nozzle?

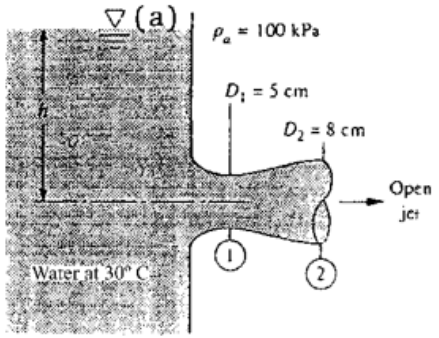


Fig. P3.133

Solution: Applying Bernoulli from (a) to (2) gives Torricelli's relation: $V_2 = \sqrt{(2gh)}$. Also,

$$V_1 = V_2(D_2/D_1)^2 = V_2(8/5)^2 = 2.56V_2$$

Vapor bubbles form when p_1 reaches the vapor pressure at 30°C, $p_{vap} \approx 4242$ Pa (from Table A.5), while $\rho \approx 996$ kg/m^3 at 30°C (Table A.1). Apply Bernoulli between 1 and 2:

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1 \approx \frac{p_2}{\rho} + \frac{V_2^2}{2} + gz_2, \quad \text{or:} \quad \frac{4242}{996} + \frac{(2.56V_2)^2}{2} + 0 \approx \frac{100000}{996} + \frac{V_2^2}{2} + 0$$

Solve for $V_2^2 = 34.62 = 2gh$, or $h = 34.62/[2(9.81)] \approx \mathbf{1.76\ m}$ *Ans.*

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Practice Problem Solution

3.126 Given a steady isothermal flow of water at 20°C through the device in Fig. P3.126. Heat-transfer, gravity, and temperature effects are negligible. Known data are $D1 = 9$ cm, $Q1 = 220$ m^3/h, $p1 = 150$ kPa, $D2 = 7$ cm, $Q2 = 100$ m^3/h, $p2 = 225$ kPa, $D3 = 4$ cm, and $p3 = 265$ kPa. Compute the rate of shaft work done for this device and its direction.

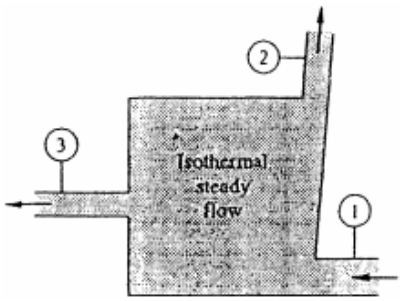


Fig. P3.126

Solution: For continuity, $Q3 = Q1 - Q2 = 120$ m^3/hr. Establish the velocities at each port:

$$V_1 = \frac{Q_1}{A_1} = \frac{220/3600}{\pi(0.045)^2} = 9.61\ \frac{m}{s};\quad V_2 = \frac{100/3600}{\pi(0.035)^2} = 7.22\ \frac{m}{s};\quad V_3 = \frac{120/3600}{\pi(0.02)^2} = 26.5\ \frac{m}{s}$$

With gravity and heat transfer and internal energy neglected, the energy equation becomes

$$\dot{Q} - \dot{W}_s - \dot{W}_v = \dot{m}_3\left(\frac{p_3}{\rho_3} + \frac{V_3^2}{2}\right) + \dot{m}_2\left(\frac{p_2}{\rho_2} + \frac{V_2^2}{2}\right) - \dot{m}_1\left(\frac{p_1}{\rho_1} + \frac{V_1^2}{2}\right),$$

1. 2.

$$\text{or:}\quad -\dot{W}_s/\rho = \frac{100}{3600}\left[\frac{225000}{998} + \frac{(7.22)^2}{2}\right] + \frac{120}{3600}\left[\frac{265000}{998} + \frac{(26.5)^2}{2}\right] + \frac{220}{3600}\left[\frac{150000}{998} + \frac{(9.61)^2}{2}\right]$$

Solve for the shaft work: $\dot{W}_s = 998(-6.99 - 20.56 + 12.00) \approx \mathbf{-15500\ W}$ *Ans.*

(negative denotes work done *on* the fluid)

Notes:

- This question is one example when it's simpler to use the partially derived equation (3.69) than the fully derived Energy conservation equation.

From Textbook:

$$\dot{Q} - \dot{W}_s - \dot{W}_v = -\dot{m}_1(\hat{h}_1 + \tfrac{1}{2}V_1^2 + gz_1) + \dot{m}_2(\hat{h}_2 + \tfrac{1}{2}V_2^2 + gz_2) \qquad (3.69)$$

Two energy terms become zero, they are crossed out and numbered in red from the solution, explanation:

1. Recall $\dot{Q}$ on the left-hand side is rate of heat transfer (not to be confused with flow rate Q)

2. The question assumes inviscid flow (no friction) and therefore work done by viscid stress is zero ($W_v = 0$)

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